VISHWA KUMAR VENKATESWARAN

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SUMMARY

Security researcher with 10+ years of programming experience (C/C++, Python, Java) specializing in vulnerability research, exploit development, and reverse engineering. Active bug bounty hunter across Bugcrowd/HackerOne/VDP programs, with hands-on work in memory-corruption and protocol-level analysis, Windows internals, malware/RE tooling, and building offensive-security infrastructure end-to-end — recon, exploitation, PoC, and disclosure.

TECHNICAL SKILLS

- **Languages:** C/C++, Python, Java, Bash, C# (.NET)
- **Automation, Software & Systems:** Data Structures & Algorithms, OOP, Docker, Linux, Git, REST/Relational Data Flows, n8n
- **Security & Vulnerability Research:** Burp Suite, Metasploit, Ghidra, Nmap, Wireshark, YARA, static/dynamic analysis
- **Automation & Detection Engineering:** Wazuh, CyberChef, Python-based automation pipelines
- **AI & Vibe Coding Tools:** Ollama, Qwen, Hugging Face, TensorFlow, LangChain4j, OpenClaw, Hermes, Cursor, Codex, Replit, Lovable

EXPERIENCE

Bug Bounty & Vulnerability Researcher — Bugcrowd *(Remote)* *May 2026 – Present*

- Perform independent vulnerability research on production web applications, identifying and responsibly disclosing security flaws through Bug Bounty and VDP channels.
- Design and run structured reconnaissance and manual testing methodologies to validate findings before submission, reducing false-positive reports.
- Write up and document root-cause analysis for each finding, following coordinated disclosure best practices used across enterprise security teams.

Summer Intern — Wimera Systems*, Bengaluru (Onsite)* *May 2026 – Jun 2026*

- Built and validated Modbus RTU and TCP/IP communication layers between industrial devices and software systems, supporting 32+ concurrent device connections with <200ms latency.
- Shipped 2+ C# (.NET) tools for protocol validation and device communication testing, cutting manual troubleshooting time on the team.
- Debugged 50+ protocol/system-level failures across a distributed set of connected devices, strengthening reliability of the communication stack.
- Implemented multi-register read support (Modbus FC03/FC04), a direct exercise in optimizing throughput under concurrency constraints.

Machine Learning Intern — SaiKet Systems *(Remote)* *Jan 2026 – Feb 2026*

- Applied ML techniques for classification and anomaly detection on security-related datasets, including data preprocessing and model optimization.
- Documented experimental workflows for reproducibility, a practice carried into later independent research work.

Security Intern — InLighnX Global Pvt. Ltd. *(Remote)* *Oct 2025 – Dec 2025*

- Completed structured training in offensive and defensive security methodologies, then built Python automation tools for authentication testing and file protection.
- Applied hashing (MD5/SHA) and password-cracking techniques to evaluate real-world credential handling weaknesses.
- Ran supervised vulnerability assessments and penetration-testing exercises against target systems.

PROJECTS

USB Rubber Ducky *Jun 2026 – Aug 2026*

- Developed a Raspberry Pi Pico–based USB HID attack simulation platform to assess endpoint exposure to malicious USB devices.
- Simulated keystroke-injection attacks in an isolated lab environment to evaluate endpoint security controls and attack detection capabilities.
- Conducted security assessments focused on unauthorized HID devices, USB-based initial access, and endpoint hardening.

AI SOC Lab — Autonomous Detection & Response Platform *May 2026 – Jun 2026*

- Engineered a distributed Wazuh + Ollama + Python detection pipeline across 15+ simulated attack scenarios, reaching 92% alert accuracy and <5min mean time to respond.
- Automated YARA rule generation and threat-hunting workflows, cutting manual log analysis time 60% (50 hrs → 20 hrs per 1M events) — a direct exercise in automation and scale.
- Deployed containerized (Docker/Linux) lab infrastructure for real-world EDR simulation and validated detection logic against MITRE ATT&CK TTP mappings.
- Analyzed attack behavior and investigated defensive controls including device allowlisting, USB restrictions, endpoint monitoring, and user-awareness measures.

Phishing Simulation Research Tool *Jan 2026 – Apr 2026*

- Built a phishing simulation toolkit for controlled security research and awareness testing, with tunnel-based hosting and customizable web templates.
- Researched browser permission handling and client-side attack surfaces to identify realistic social-engineering vectors.
- Applied ethical security research practices throughout, mirroring how enterprise red teams validate user-facing attack paths.
- Automated phishing campaign setup and response tracking to evaluate user interaction and security awareness.

Multi-Agent Banking Assistant *Dec 2025 – Feb 2026*

- Engineered a privacy-preserving multi-agent banking system using LangChain4j, separating intent classification from privileged banking operations.
- Integrated local LLaMA 3 inference via Ollama to minimize exposure of sensitive financial data to third-party AI services.
- Implemented a controlled LLM-to-API tool-calling architecture with domain-specific agents.
- Built a React interface for document ingestion and AI-assisted financial workflows,

ACHIEVEMENTS & RECOGNITION

- Certified: Android Bug Bounty Hunting (EC-Council); BASH Training (IIT Bombay).
- Active CTF competitor — 10+ HackTheBox machines completed; top 15% on TryHackMe.
- Publishes technical security research and CTF writeups: vishwakumarv.github.io/writeups

EDUCATION

PSG College of Technology, Coimbatore *Aug 2023 – May 2027 (Expected)*

Bachelor of Engineering in Electronics and Communication Engineering